C++ Programming

Trainer : Rajiv K

Email: rajiv.kamune@sunbeaminfo.com

Agenda

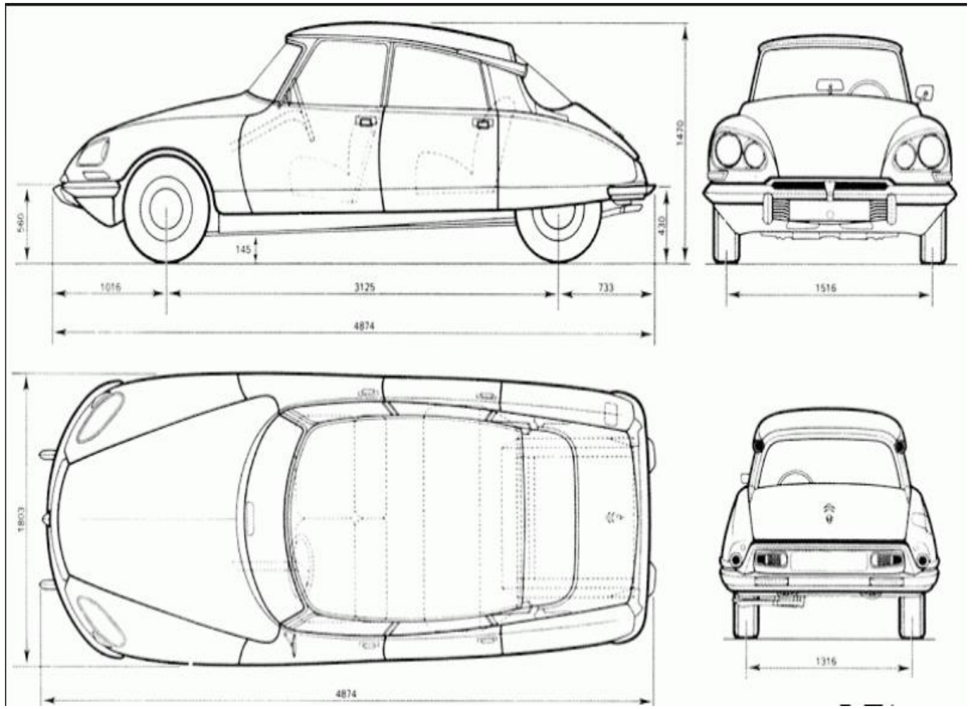
- class and object
- Data Member & Member Functions
- Types of Member Functions
- Constructor
- Facilitator
- Inspector and Mutator
- Destructor
- this pointer
- Constructor initializer list
- Exception Handling

class and object

- Everything in C++ is associated with **classes and objects**, along with its properties and behavior.
- The class is **template** for **an Object**
- Or you may say it is a **blue print for an object**, by looking towards which an object gets instantiated.
- Class is collection of data member and member function.
- Object is an **instance** of a class.
- Entity that has **physical existence**, **can store data**, **send and receive message** to communicate with other objects.

Example of Class and Object

Blue Print of Car



Objects of a Class Car

class and object

- Example:

Blue Print of building

Real life entity(Object)

Object

- An entity, which **get space inside memory** is called object.
- Object is used to access **data members** and **member function** of the class
- Process of creating object from a class is called **instantiation**
- **Object has**
  - Data members (***state*** of object)
    - Value stored inside object is called **state of the object**.
    - Value of data member represent **state of the object**.

  - Member function (***behavior*** of object)
    - Set of operation that we perform on object is called behaviour of an object.
    - Member function of class represent behaviour of the object.
    - is how object acts & reacts, when its state is changed & operations are done
    - Operations performed are also known as messages

Object

- **State**: State of an object contains all the **properties** of the object and **values of** each and every the **properties**.

- **Value Stored in side an object** is called state of an object.

- **Behavior:** Behavior is how an object **act and react**, when state of an object changes and function gets called.

- **Member function** of a class is nothing but the **behavior** of an object.
- **Identity**: Identity is **the property of a class** by which it gets **differentiated** from all other objects.

Data Member & Member Functions

- Unique address(***identity*** of object)
  - Value of any data member, which is used to identify object uniquly is called its identity.
  - If state of object is same then its **address** can be considered as its identity.

- **Data members are the data variables and member functions are the functions used to manipulate these variables** and together these data members and member functions defines the properties and behavior of the objects in a Class.
- The data members and member functions of class can be accessed using the dot('.')

Types of Member Functions

- Constructor
- Facilitator
- Inspector
- Mutator
- Destructor
- **Constructor:**
  - Constructor is a **special member** function of a class **having same name** as a class and having no return type.
  - Constructor is used to **initialize an object/data member** of a class.
  - In entire **life cycle** of an object constructor **gets called only once**.

Types of Constructor

- **Types of Constructor:**
  - 1. Parameterless Constructor
  - 2. Parameterised Consructor
  - 3. Default Constructor
- **1. Parameterless Constructor:**
  - Parametrless constructor is a constructor which **does not have parameter** is called as parameterless constructor.
  - It is also called as **zero parameter constructor** of user defined default contructor.
- **2. Parameterised Consructor:**
  - Parameterised constructor is a constructor which **dose have one or more parameter** is called as parameterless constructor.
- **3. Default Constructor:**
  - If no constructors are explicitly declared in the class, a default constructor is **provided automatically by the compiler**

Types of Member Functions

- **Facilitator:**
  - Causse an object to perform some action or service.
- **Inspector:**
  - public member functions in a class that get (accessors) the value of data member of a class.
- **Mutator:**
  - a mutator method is a method used to **change the value** of a data member of the class.

Types of Member Functions

- **Destructor:**
  - Destructor is also a special member function like constructor.
  - An operation that **deinitializes an object.**
  - A member function that is **invoked automatically** when the **object goes out of scope or is** explicitly destroyed by a call to **delete**.
  - Destructor neither **requires any argument nor returns any value.**

This Pointer

- this is a **key word** in C++
- It is **implicit constant pointer** available in side the each non static member function of a class.
- This pointer stores the address of current object of a class.
- If **name** of **data member** and **parameter passed** to **the member function** is **same** then it becomes **mandatory** to use **this pointer** inside that function to differentiate between the data member and local parameter to the function.
- Following are the function which doesn't get this pointer
    - Global Function
    - Static Function
    - Friend Function

Constructor's member initializer list

- If we want to initialize data members according to users requirement then we should use constructor body.

```
class Test
{
private:
        int num1;
        int num2;
        int num3;
public:
        Test( void )
        {
        this->num1 = 10;
        this->num2 = 20;
        this->num3 = num2;
        }
};
```

- If we want to initialize data member according to order of data member declaration then we can use **constructors member initializer** list.

```
class Test
{
private:
        int num1;
        int num2;
        int num3;
public:
        Test( void ) : num1( 10 ), num2( 20 ), num3( num2)
        { }
};
```

Except array we can initialize any member inside constructors member initializer list.

Empty Class

Empty Class :-

A class which **do not have any Data member or member function** is called as the Empty class.

- We **can create an object** of a empty class.
- **Size** of an object of a empty class is **1 byte**.

Difference between class and Structure in C++

Class	Structure
Members of a class are private by default.	Members of a structure are public by default.
Member classes/structures of a class are private by default.	Member classes/structures of a structure are public by default.
It is declared using the **class** keyword.	It is declared using the **struct** keyword.
It is normally used for data abstraction and further inheritance.	It is normally used for the grouping of data

Exception Handling

- One of the **advantages of C++ over C** is Exception Handling. Exceptions are **runtime anomalies/error** or **abnormal conditions** that a program encounters during its execution.

- Exception is an **event or object** which is thrown at runtime.

- **Exception Handling** in C++ is a **process to handle runtime errors**. We perform exception handling so the **normal flow** of the application can be **maintained even after runtime errors**.

- Exception need to be handled other wise it would terminate the program abnormally.

Exception Handeling:

- To create an application, most of the times we take help of operating system resources. Following are some the resources we use frequently.
  - 1. Memory
  - 2. File
  - 3. Socket etc.
- In C++ we can handle exceptions with the help of following keywords:
  - 1.try
  - 2.catch
  - 3.throw

Exception Handling:

- **1. try block/ try handler:**
  - if we want to **inspect** group of statements for **exception** then we should put such statement **inside try block** or **try handler**.
  - **Syntax:**

```
try{
        code statement 1;
        code statement 2;
        code statement 3;
    }
```

Code which may possibly generate the exception

Exception Handling:

- **2. catch block/catch handler:**
  - If we want to **use try** block then it is **necessary** to provide at **least one catch** block.
  - To **handle exception** thrown **from try block** we should **use catch block** or catch handler. Single try block may have multiple catch block but it must have at least one catch block.
  - **Syntax:**

```
try{
    code statement 1;
    code statement 2;       //Code which may possibly generate the exception
        |
        |
    code statement n;
    }
catch()
    {
        //Block of code to handle the error
    }
```

Exception Handling

- **Generic Catch block/Generic catch handler:**
  - A catch block which can handle all types of exception is called generic catch block.
  - In case if you don't have an idea which error will be thrown by the block of code then in such case you need to give generic block for the try.
  - **Syntax:**

```
try{
    code statement 1;
    code statement 2;
          |
          |
    code statement n;
    }
catch(…)
    {
        //Block of code to handle the error
    }
```

//Code which may possibly generate the unknown exception

Exception Handling

- **3. throw:** A program throws an exception when a problem shows up. This is done using a **throw** keyword.
- **Example:**

```
try
{
     cout<<“Before throw…!”;
     throw 10;
     cout<<“After throw…!”
}
Catch(int n)
{
     cout<<“Error code:”<<n;
}
```

Thank You